Introduction to Quest

Northwestern IT - Research Computing Support

June 30, 2026

Northwestern

Overview

1. What is High-Performance Computing?
2. Make-up of Quest
3. SLURM Job Scheduler
4. Types of Jobs
   - Submit a batch job to Quest
   - Submit a Quest OnDemand job
5. Checking on Jobs with Slurm
6. File transfers and Storage Practices
7. Software on Quest
8. TLDR

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High-Performance Computing

HPC uses supercomputers and computer clusters to solve advanced computation problems. – Wikipedia

HPC encompasses solutions that are able to process data and execute calculations at a rate that far exceeds other computers. This aggregate computing power enables [scientists] to solve large problems that would otherwise be unapproachable. – HPE

HPC is technology that uses clusters of powerful processors, working in parallel, to process massive multi-dimensional datasets (big data) and solve complex problems at extremely high speeds. - IBM

When Do You Use an HPC?

- Training Large AI Models
  - Models have billions of parameters
  - Training requires huge processing datasets
  - Needs GPUs/accelerators working in parallel

- Genomics and DNA Sequencing
  - Human genome = ~3 billion base pairs
  - Modern sequencing produces terabytes of data per experiments

- Engineering Simulations
  - Millions of elements modeled in fine detail
  - Requires solving large systems of equations repeatedly

When Do You Use an HPC?

- Doing this on a laptop:
  - Not enough disk space
  - Not enough processors
  - Not enough time

HPC at Northwestern - Quest Computing Cluster

- Provides computing for all researchers from a wide range of fields
- Enables the use of multiple powerful computers simultaneously
- 5,500 active users, 1231 compute nodes, 80,390 CPU cores, 304 GPU Cards, 750 research software, 78 knowledge base articles, 28 training recordings

Quest

Other HPC clusters

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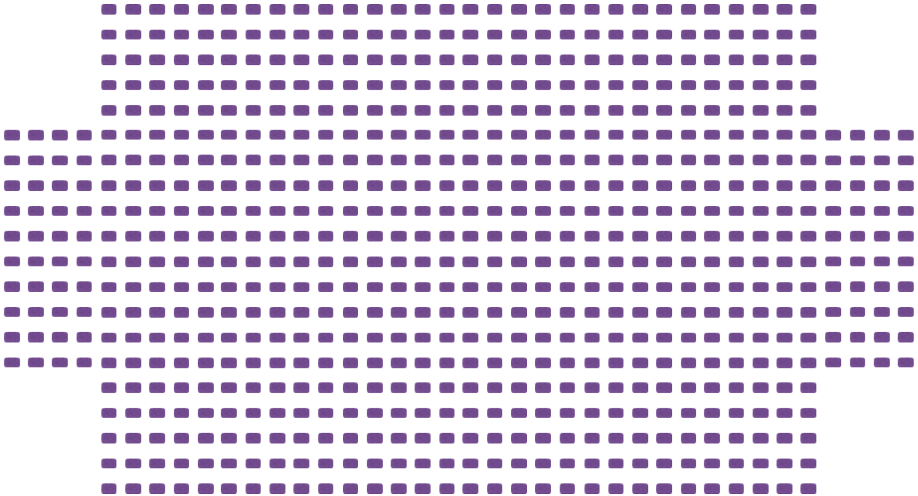
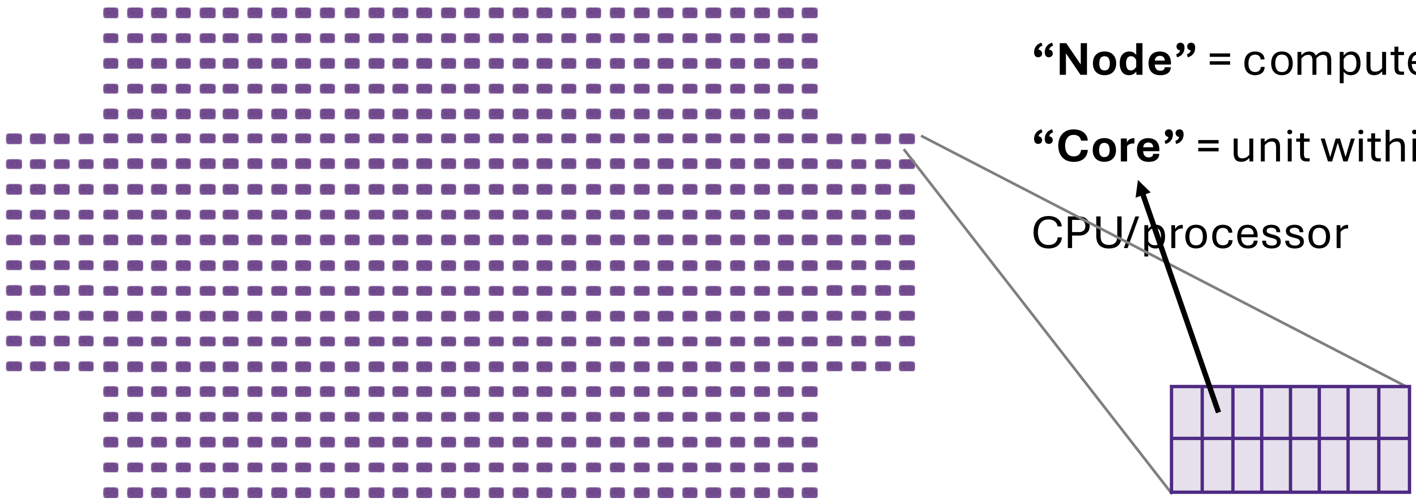
Quest consists of ~1200 nodes



“Node” = computer

Each node consists of 52-128 cores

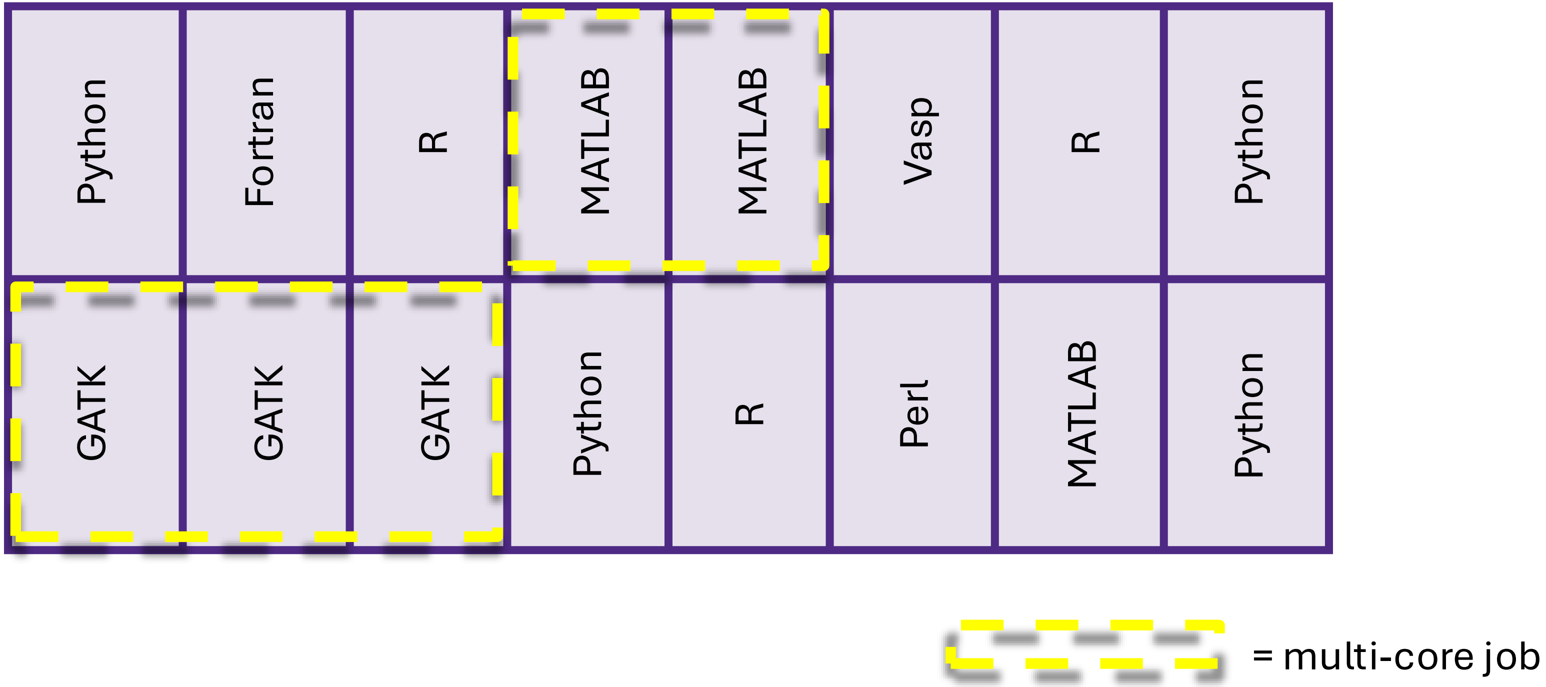


“Node” = computer

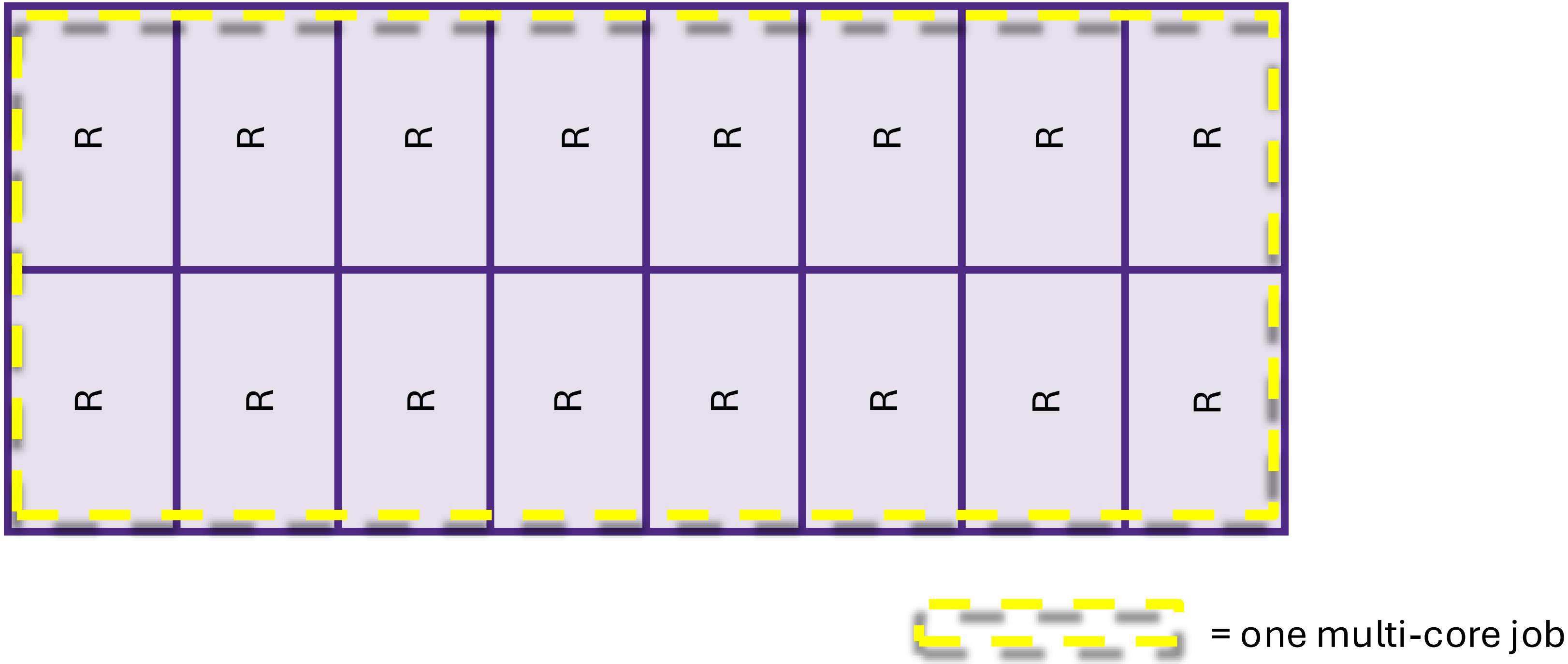
“Core” = unit within CPU/processor

“Memory (RAM)” = Very fast temporary storage for processors

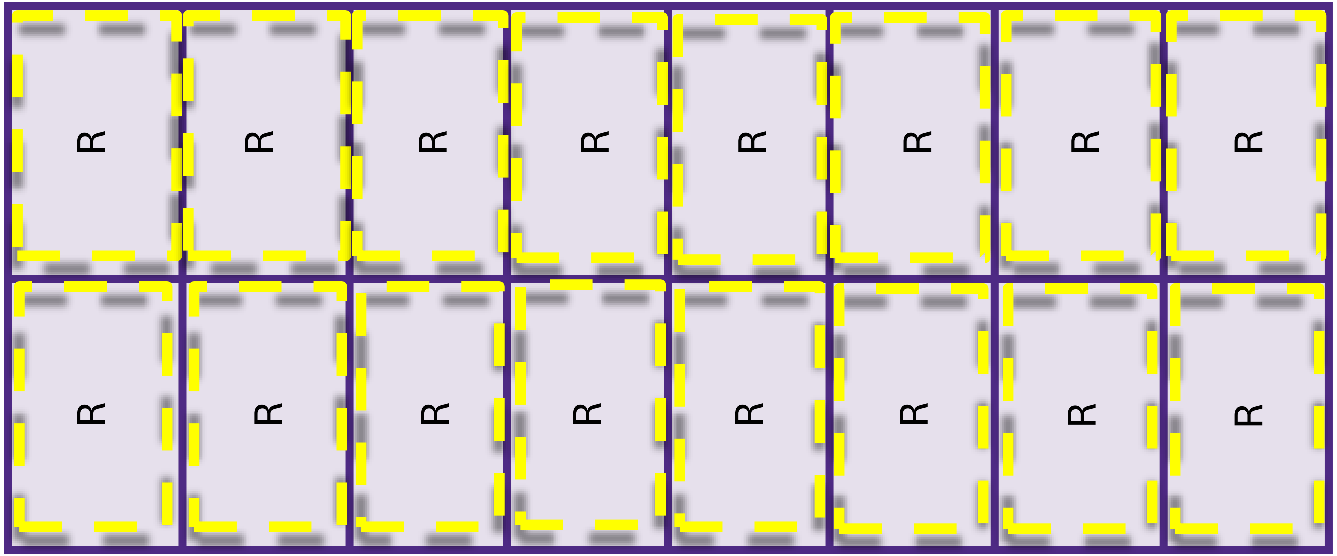
Each node can run separate jobs on each core



A node can run a job with 52-128 cores at a time

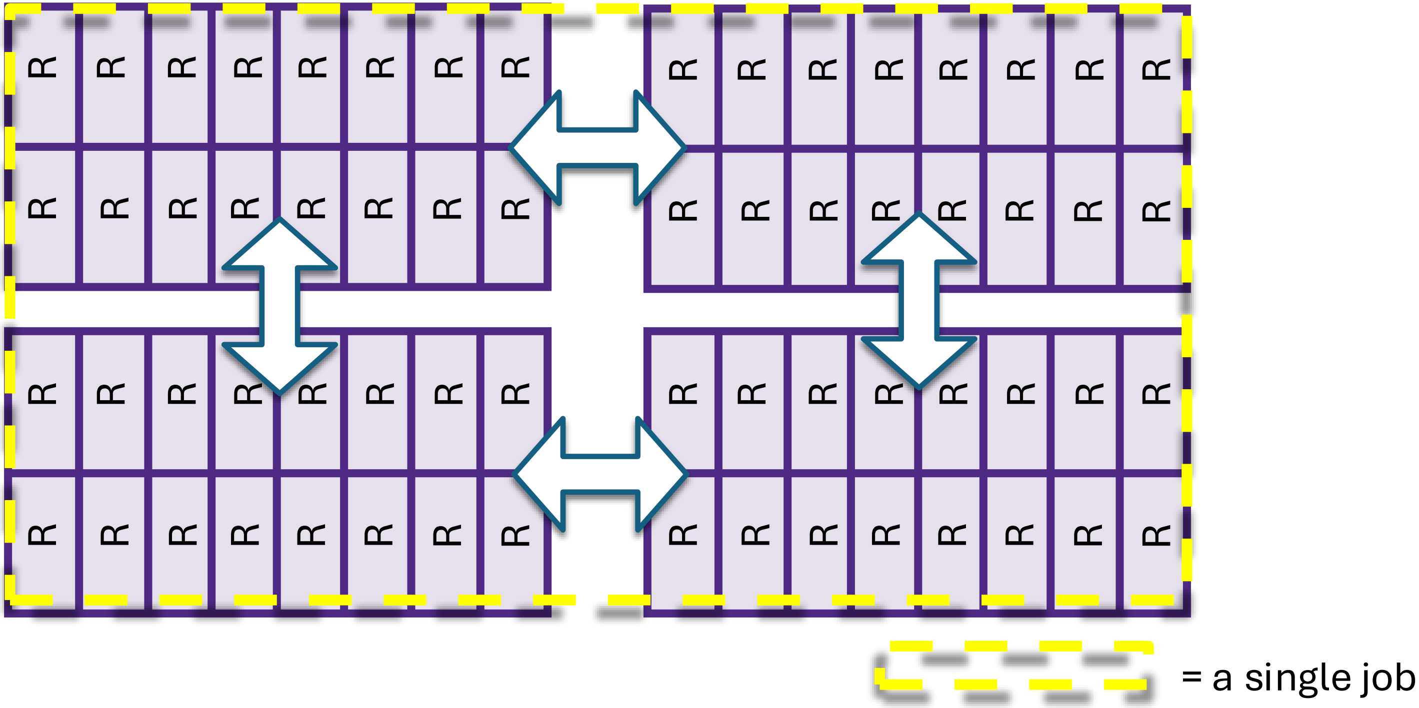


You can run thousands of jobs at a time



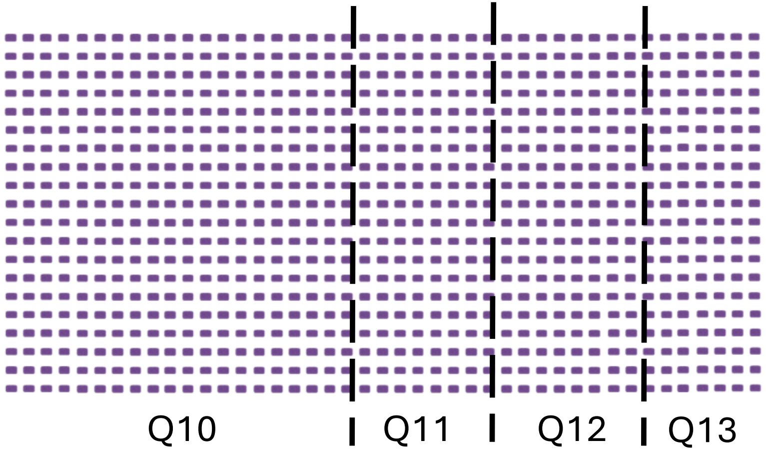

= a single job

Jobs can even run across multiple nodes



Technical specifications of nodes vary

There are several generations of nodes!

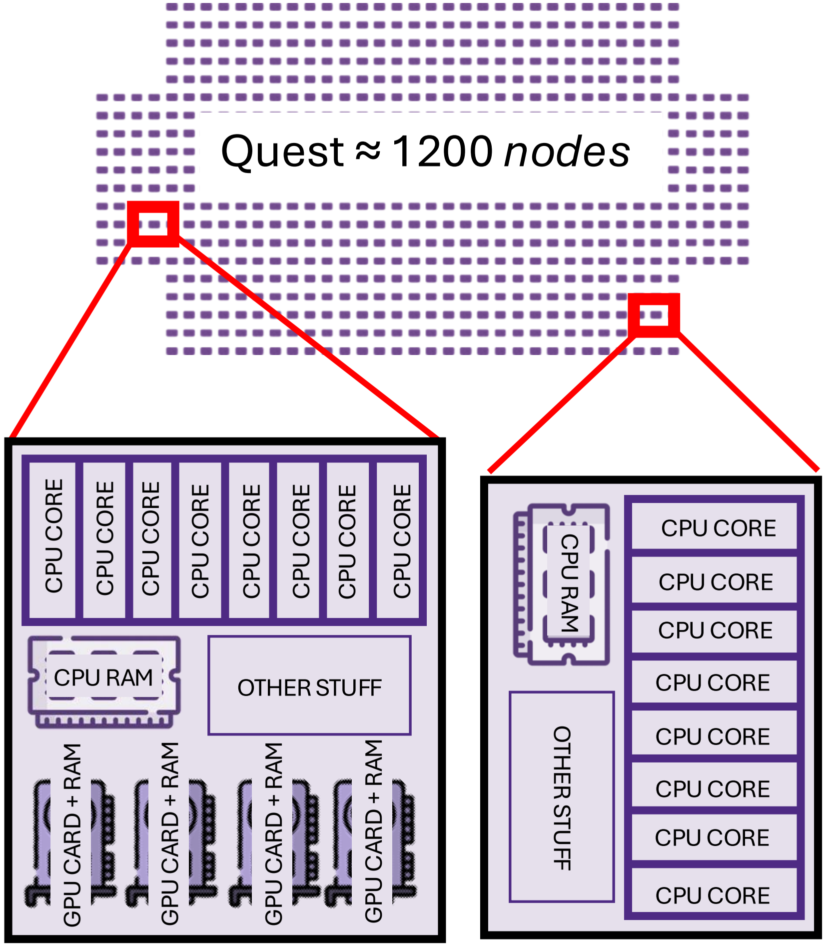


Generation	# nodes	# cores per node	Total usable memory (RAM)	Memory per core
Q10	555	52	166 GB	~3.2 GB
Q11	209	64	221 GB	~3.5 GB
Q12	214	64	221 GB	~3.5 GB
Q13	140	128	473 GB	~3.7GB
Your laptop (probably)	1	10-25	16-128GB	~2-8GB

Computing Resources

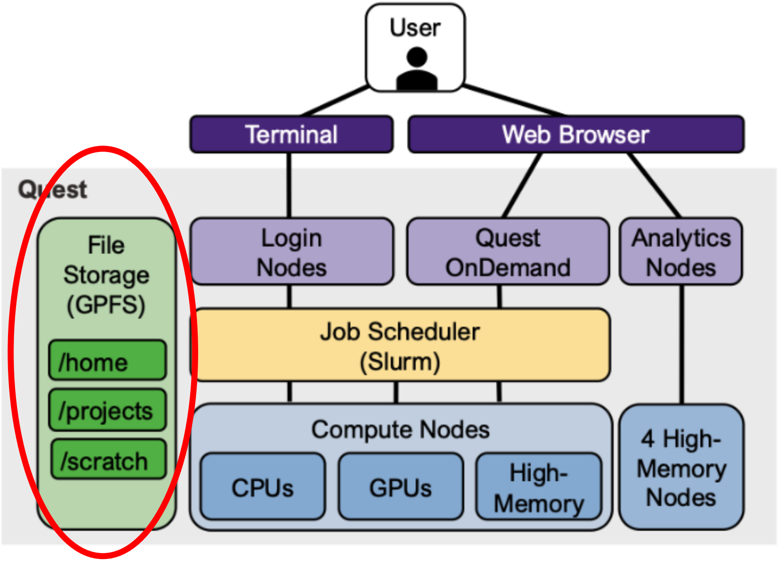
Nodes have *schedulable resources*:

- 52-128 *CPU cores*
- 192 - 1500 GB of *CPU RAM* (total)
- 0 - 4 *GPU cards*

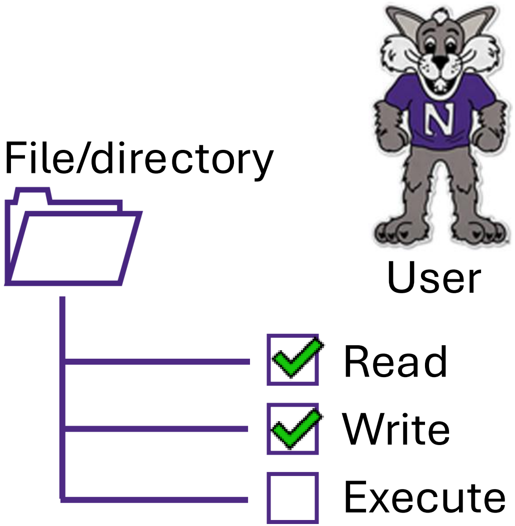


Quest Filesystem and Working Directories

- `/home` - all users
  - 80 GB
  - Backed up
- `/projects` - all allocations
  - 1 or 2 TB
  - Not backed up
- `/scratch` - upon request
  - 5 TB
  - Removed after 30 days



What are “permissions”?



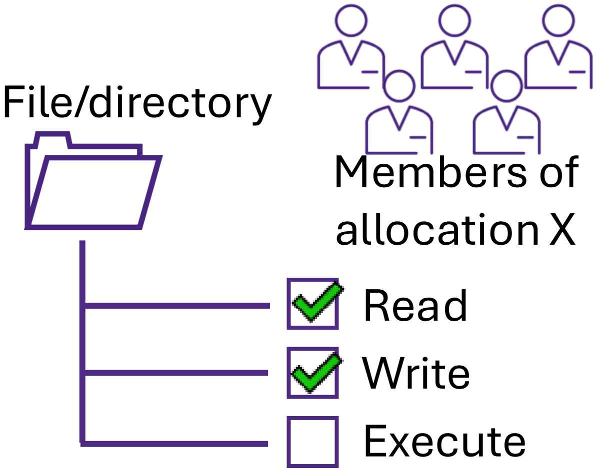
Files you created:

- You will always have full read/write permissions.

Files you did not create:

- You may not have read/write permissions.

Group-level permissions



Group permissions

- Permission are also defined at the “user group” level.
- To see which user groups you are a part of, run “groups”

```
[netid@quser21 ~]$ groups
netid     p30XXX     b10XX
```

Who will have access to my files?

Files created in /projects

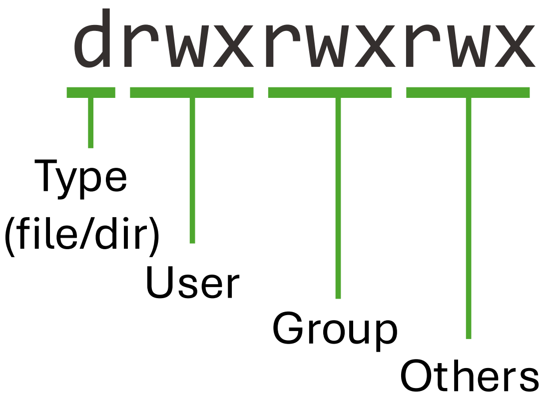
- Read/write permissions to all allocation members by default

Files created in /home

- Read/Write permissions only you by default

Permission strings

```
[abc1234@quser21 p12345]$ ls -l
total 2
drwxrwxr-x 1 abc1234 p12345 4096 Apr 15 11:00 codes
drwxrwxr-x 1 abc1234 p12345 4096 Apr 15 10:00 data
-rw-rw-r-- 1 abc1234 p12345  259 Apr 15 09:00 test.txt
```



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Job Scheduler

- Job Scheduler = software that manages how computing resources are allocated to users and allocations on an HPC system

- Instead of users running programs directly on compute nodes, they submit jobs to the scheduler, which determines where, when, and how these jobs will run.

- Quest’s job scheduler is SLURM
    - Simple Linux Utility for Resource Management

Job Scheduler

Order of operations:

1. User submits a job script
2. Scheduler places the job in a queue
3. Scheduler evaluates priorities and resource availability
4. Resources are allocated
5. Jobs runs on the assigned compute nodes
6. The results are returned and the resources are released

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Job Types

- Batch job
  - Jobs that are submitted via script
  - Good for: code that has a longer runtime/more resources
- Interactive job
  - Jobs that have a graphical user interface (GUI), or real-time shell access
  - Good for: debugging and prototyping

SBATCH Example

```
#!/bin/bash
#SBATCH --account=p12345
#SBATCH --partition=short
#SBATCH --nodes=1
#SBATCH --ntasks-per-node=1
#SBATCH --time=00:10:00
#SBATCH --mem=1G
#SBATCH --job-name=sample_job
#SBATCH --output=output.log

module purge all
module load python-anaconda3
source activate /projects/intro/envs/slurm-py37-test

python --version
python slurm_test.py
```

sbatch headers

Setup software environment

Run your code

Job Submission and Monitoring

- Submit a job to the scheduler to run
  - `sbatch submission_script.sh`
- How do I check if my job is running?
  - `squeue -u <NETID>`

Job Resources

- Quest partitions:
  - `short | normal | long | gengpu`

- More memory | CPUs | GPUs | = longer queue wait time

- Make sure your code can use the resources:
  - Multi-CPU
  - Multi-GPU

Exercise: submit a batch job to run on Quest

- Navigate to your folder in `/projects/e53006`
- Clone the git repository
  https://github.com/nuitrcs/intro_quest_reu
- Use a command line text editor to edit the submission script
  `submission_script.sh`
- Submit the job to run on Quest
- Examine the output

Exercise: submit a batch job to run on Quest

```
[abc1234@quser41 ~]$ cd /projects/e53006/<first_last>

[abc1234@quser41 ~]$ git clone
https://github.com/nuitrcs/intro_quest_reu.git

[abc1234@quser41 ~]$ cd intro_quest_reu

[abc1234@quser41 ~]$ nano submission_script.sh

[abc1234@quser41 ~]$ sbatch submission_script.sh

[abc1234@quser41 ~]$ squeue -u <NETID>

[abc1234@quser41 ~]$ cat slurm-<JOBID>.out
```

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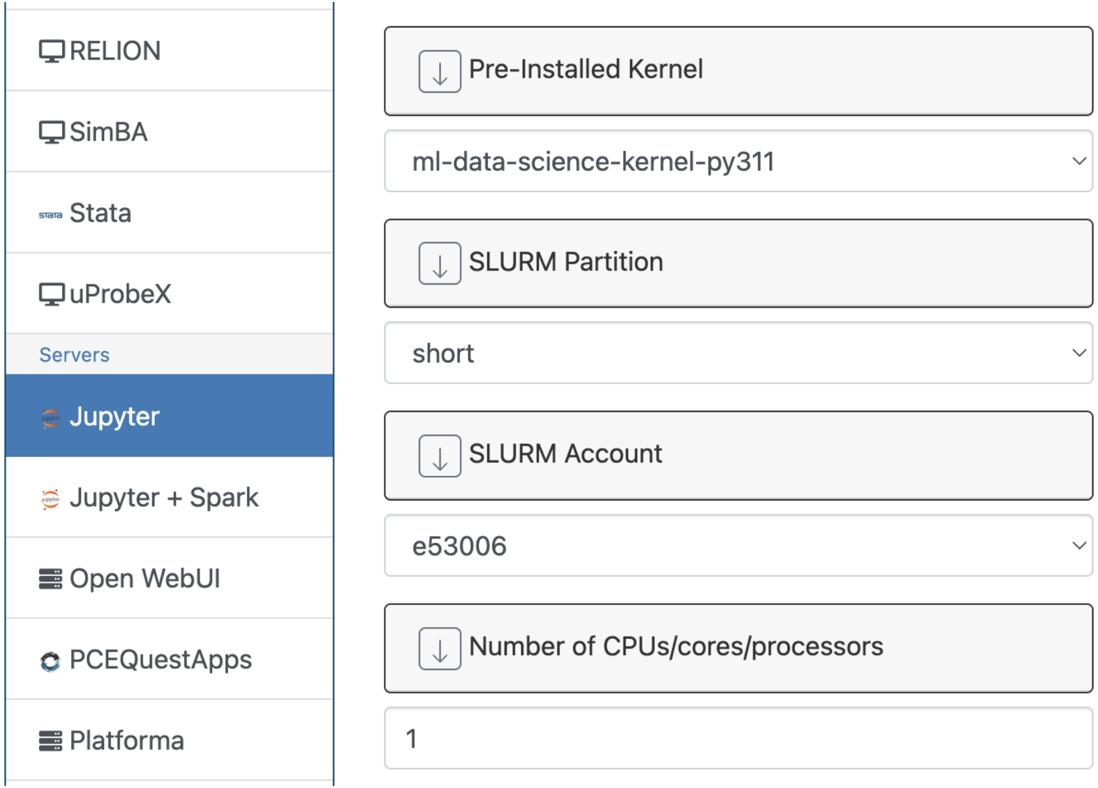
Quest OnDemand

- Open OnDemand
  - Good for interactive development!
  - Requires VPN
  - Uses Quest partitions
- https://ondemand.quest.northwestern.edu/

Exercise: submit a Quest OnDemand job

- Navigate to https://ondemand.quest.northwestern.edu/
- Select the Jupyter Interactive Application
- Complete the job submission form using:
  - SLURM partition = `short`
  - SLURM account = `e53006`
  - Jupyter root directory = `/projects/e53006/`
- Run the notebook `happy_face.ipynb` in your clone of the `intro_quest_reu github` repository with the kernel `ml-data-science-kernel-py311`

Exercise: submit a Quest OnDemand job



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Checking running jobs: squeue

- `squeue -u <NETID>`
  - Shows the jobs in the queue for the given user.
  - Queued jobs have the status Pending, or Running.
  - Pending jobs will list the reason they are waiting
  - Resources means you are next in line but the resources aren’t available
  - Priority means a different job in the queue has higher priority

Checking job history: sacct

- `sacct -X -S <startdate>`
    - Shows your job history.
    - Default to only today's history if no start date is provided.
    - Start dates should be provided in MMDDYY.
    - This is a good way to look up Job ID numbers.

Checking job resource use: seff / checkjob

- `seff <jobid>`
  - Shows a job’s resource use efficiency
  - Only accurate after a job has completed
- `checkjob <jobid>`
  - Shows job details including, seff output
  - Also includes Slurm variables and the job script

Exercise: look at your Quest job history

- Try running the following commands:
- `sacct -X`
- `sacct -X --format=jobid,jobname%20,account,partition,submit, elapsed,state,nodelist`
- `seff <ONE_OF_YOUR_RECENT_JOBIDS>`
- `checkjob <ONE_OF_YOUR_RECENT_JOBIDS>`

Exercise: look at your Quest job history

```
[abc1234@quser41 ~]$  sacct -X
JobID             JobName  Partition    Account  AllocCPUS      State  ExitCode
------------ ---------- ---------- ---------- ---------- ---------- --------
5499423       my_test_j+      short      e53006          1  COMPLETED      0:0
5499621       jupyter_j+      short      e53006          1 CANCELLED+      0:0

[abc1234@quser41 ~]$  sacct -X --format=jobid,jobname%20,account,partition,submit,elapsed,state,nodelist
## Output is long ##

[abc1234@quser41 ~]$  seff 5499423
Job ID: 5499423
Cluster: quest
User/Group: tdm5510/tdm5510
State: COMPLETED (exit code 0)
Cores: 1
CPU Utilized: 00:00:02
CPU Efficiency: 7.41% of 00:00:27 core-walltime
Job Wall-clock time: 00:00:27
Memory Utilized: 16.05 MB
Memory Efficiency: 0.39% of 4.00 GB (4.00 GB/node)

[abc1234@quser41 ~]$  checkjob 5499423
## Output is long ##
```

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Transferring Files

- What's the best way to move data to Quest?
  - Globus!
- Quest, RDSS, and NU Sharepoint are Globus endpoints
- Command line options

```
alice@laptop ~ % scp file.txt alice@quest.login.northwestern.edu:/path/to/destination/
```

file to transfer from source username and remote hostname destination path

Storage Best Practices

- Don't have more than 7,500 files in a single directory
- In aggregate, don't have more than 50 million files in a full directory path
- Helpful utilities
  - `homedu`
  - `checkproject`
  - `dust` (module)

Storage Best Practices

- Quest cannot store data subject to compliance frameworks (HIPAA, FERPA, NIST 800-171, NIST 800-53, FISMA, FedRAMP, etc.)
  - Contact quest-help@northwestern.edu with any questions

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Software on Quest

- Manage Python and R packages through virtual environments
- Can I compile my own applications?
  - Yes, you are responsible for understanding and obeying license requirements
- Can the RCS team help me compile an application?
  - Yes!
  - Submit a ticket to quest-help@northwestern.edu

Virtual Environments

- Access Python on R packages from within jobs submitted to the scheduler
- Improve portability and reproducibility
- Isolate dependencies per project
- Python
  - Use Conda or Mamba
- R
  - Use renv

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Summary

- Quest (and HPC clusters in general) provides computation and storage resources for research workflows
  - More CPUs, memory, and storage than a personal laptop or workstation
- Be a conscientious user
  - Strive for efficient resource usage
  - Start small and prototype/test codes before scaling up
  - Design your software environments and code repositories with reproducibility in mind
- Reference our documentation and/or reach out to quest-help@northwestern.edu for help

Thank you!

Appendix

Quest Utilities

- groups - list your allocations
- pwd - primary working directory
- ls - list contents of directory
- cd - change directories
- homedu - disk usage in your home directory
- checkproject - disk usage and expiry of an allocation

Quest Utilities

- cp - copy file(s) from one location to another
- mv - move a file from one location to another
  - Will delete the file from original location!
- rm - remove a file
  - Can't be undone!
- cat - print the contents of a file
- head - print the first part of a file
- nano - open a file with the command line text editor nano

Job Submissions

- How do I know which partition to use?
  - short | normal | long | gengpu
- How do I check my job?
  - sacct or squeue
- How long before my job will be scheduled?
  - Depends on how many people are using Quest and the resources required

Job Submission Best Practices

- Users should not submit more than 4,700 in a single submission
- Batching jobs into groups can help to keep the cluster from getting overwhelmed
- Scale up responsibly – run a scaling test